

The General-Boundary Spectral Inequality for the Bi-Laplacian

Candidate full solution packet (GPT5.6)

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Status. Candidate full solution, likely valid, subject to expert review. The proof answers the spectral-inequality question in Section 1.6.2 of arXiv:2109.01521v2. It does not address the neighboring questions on boundary damping or quantitative boundary unique continuation.

1 Source question and later theorem

Let Ω be a smooth bounded connected domain (or a compact smooth manifold with boundary). Let B_1, B_2 be boundary differential operators of orders k_j , with normal order at most three, and let

$$\mathcal{P}u = \Delta^2 u, \quad \mathcal{D}(\mathcal{P}) = \{u \in H^4(\Omega) : B_1 u = B_2 u = 0 \text{ on } \partial\Omega\}.$$

The standing hypotheses in Le Rousseau–Zongo [1] are:

1. (Δ^2, B_1, B_2) satisfies the Lopatinskii–Shapiro condition;
2. $\mathcal{P}$ is self-adjoint and nonnegative on $L^2(\Omega)$.

Writing (ϕ_j) for an orthonormal eigenbasis and (μ_j) for the eigenvalues, Section 1.6.2 asks whether, for every nonempty open $\omega \subset \Omega$, there is $C > 0$ such that

$$\|u\|_{L^2(\Omega)} \leq C e^{C\mu^{1/4}} \|u\|_{L^2(\omega)}, \quad u \in \text{span}\{\phi_j : \mu_j \leq \mu\}. \quad (1)$$

The exact source passage is on page 8:

1.6.1. *Boundary damping.* Here, we have considered a damping that acts in the interior of the domain Ω . The study of boundary damping, as in [38] for the wave equation, is also of relevance. Yet we foresee that it requires to specify more the used boundary operators. This was not our goal here as we wished to treat general boundary operators here.

1.6.2. *Spectral inequality.* If the boundary operators B_1 and B_2 are well chosen, the bi-Laplace operator Δ^2 can be selfadjoint on $L^2(\Omega)$; see Section 9.1. Associated with the operator is then a Hilbert basis $(\phi_j)_{j \in \mathbb{N}}$ of $L^2(\Omega)$. In the case of “clamped” boundary condition the following spectral inequality was proven in [35].

Theorem 1.3 (Spectral inequality for the “clamped” bi-Laplace operator). *Let $\mathcal{O}$ be an open subset of Ω . There exists $C > 0$ such that*

$$\|u\|_{L^2(\Omega)} \leq C e^{C\mu^{1/4}} \|u\|_{L^2(\mathcal{O})}, \quad \mu > 0, \quad u \in \text{Span}\{\phi_j; \mu_j \leq \mu\}.$$

The proof of this theorem is based on a Carleman inequality for the fourth-order elliptic operator $D_s^4 + \Delta^2$, that is, after the addition of a variable s . Extending this strategy to the type of boundary conditions treated here was not successful because it is not guaranteed that having the Lopatinskiĭ-Šapiro condition for Δ^2 , B_1 , and B_2 implies that the Lopatinskiĭ-Šapiro condition holds for $D_s^4 + \Delta^2$, B_1 , and B_2 . Yet, the Lopatinskiĭ-Šapiro condition is at the heart of the proof of our Carleman estimate. Proving a spectral estimate as in the above statement for the general boundary conditions considered here is an open question.

1.6.3. *Quantification of the unique continuation property.* For a second-order operator like the Laplace operator Δ and a boundary operator B of order k (yet of order at most one in the normal direction) such that the Lopatinskiĭ-Šapiro holds one can derive the following inequality that locally quantifies the unique continuation property up to the boundary; see [33] Lemma 9.2].

Proposition 1.4. *Let $x^0 \in \partial\Omega$ and V be a neighborhood of x^0 where the Lopatinskiĭ-Šapiro condition holds. There exist W an open neighborhood of x^0 , $\varepsilon \in (0, 1)$, $\delta \in (0, 1)$, and $C > 0$, such that*

$$\|u\|_{H^1(W)} \leq C \|u\|_{H^1(V)}^{1-\delta} \left(\|\Delta u\|_{L^2(V)} + |Bu|_{H^{1-k}(V \cap \partial\Omega)} + \|u\|_{H^1(V_\varepsilon)} \right)^\delta, \quad (1.5)$$

for $u \in H^2(V)$, with $V_\varepsilon = \{x \in V; \text{dist}(x, \partial\Omega) > \varepsilon\}$.

This inequality can be obtained from a Carleman estimate as in Theorem 1.1 for the Laplace operator, yet with the large parameter τ allowed to be chosen as large as desired. This is exploited in an optimization procedure on the parameter τ in $[\tau_0, +\infty[$ for some τ_0 . Note however that in the statement of Theorem 1.1 one has $\sigma \lesssim \tau \lesssim \sigma$. Thus, the optimization procedure cannot be carried out in $[\tau_0, +\infty[$. While having a result quantifying the unique continuation under Lopatinskiĭ-Šapiro-type conditions for the bi-Laplace operator similar to that of Proposition 1.4 can be expected, the Carleman estimate we obtain in the present article cannot be used, at least directly, for a proof as carried out in [33] or in former sources such as [37].

Zongo–Robbiano [2, Theorem 1.3, p. 6] prove (1) under the preceding two hypotheses and the additional Lopatinskiĭ–Shapiro condition for

$$Q = D_s^4 + \Delta^2$$

with the same boundary operators on $(0, S_0) \times \partial\Omega$. The source paper had isolated precisely this extra condition as the obstruction: ordinary Lopatinskii–Shapiro need not imply parameter Lopatinskii–Shapiro without further structure [1, p. 8]. The point below is that the already assumed self-adjoint nonnegativity rules out the obstruction.

2 Proof intuition

The augmented boundary condition is a negative-resolvent condition in disguise. After Fourier transform in the added variable s , failure at a frequency $\sigma \neq 0$ produces a nonzero exponentially decaying solution of the frozen normal problem

$$(P_0(\xi', D_t) + \sigma^4)w = 0, \quad B_{j,0}(\xi', D_t)w(0) = 0.$$

Thus the frozen half-line realization has the negative eigenvalue $-\sigma^4$. Localizing this mode at spatial scale h gives a boundary packet with Rayleigh quotient $-\sigma^4 h^{-4} + o(h^{-4})$. The principal boundary traces vanish; the original elliptic boundary condition supplies a trace correction smaller by $h^{1/2}$. The corrected packets lie in $\mathcal{D}(\mathcal{P})$, contradicting $\mathcal{P} \geq 0$.

This proves the augmented Lopatinskii–Shapiro condition automatically. The spectral inequality is then exactly Zongo–Robbiano’s theorem.

3 The automatic parameter condition

We record the standard boundary-layer implication in the precise form needed here. Principal symbols are taken in boundary normal coordinates at a fixed $x^0 \in \partial\Omega$; $t \geq 0$ is the inward normal variable.

Lemma 1 (Trace correction at boundary-layer scale). *Assume (P, B_1, B_2) is a normal fourth-order elliptic boundary problem of the source class, satisfying ordinary Lopatinskii–Shapiro, whose realization is self-adjoint. Suppose $w(t)$ is exponentially decreasing and its principal boundary traces obey*

$$B_{j,0}(x^0, \xi', D_t)w(0) = 0, \quad j = 1, 2.$$

Let u_h be the normalized packet

$$u_h(x', x_d) = h^{-(d+1)/4} \chi(x'/\sqrt{h}) e^{ix' \cdot \xi'/h} \chi_0(x_d) w(x_d/h) \quad (2)$$

in a boundary chart, where the cutoffs equal one near the origin. Then there are smooth v_h , supported in a fixed slightly larger chart, such that

$$B_j(u_h + v_h) = 0, \quad \|v_h\|_{L^2} = O(h^{1/2}), \quad \|Pv_h\|_{L^2} = O(h^{-7/2}).$$

Proof. The normalization in (2) gives $\|u_h\|_{L^2} = \|\chi\|_2 \|w\|_2 + o(1)$ after a harmless fixed rescaling. A derivative on the carrier or on $w(x_d/h)$ costs h^{-1} , whereas a derivative on the tangential cutoff costs only $h^{-1/2}$. Freezing coefficients at x^0 therefore leaves a relative $O(h^{1/2})$ error. Since the principal B_j traces vanish, this gives

$$\|B_j u_h\|_{H^{7/2-k_j}(\partial\Omega)} = O(h^{-7/2}), \quad (3)$$

one half-power below the H^4 size $O(h^{-4})$ of the packet.

For completeness, use the boundary jet map $J = (\gamma_0, \gamma_1, \gamma_2, \gamma_3)$ and write each B_j as a tangential differential matrix acting on J . On the h^{-1} frequency annulus, the normal elliptic boundary system

gives a uniformly bounded right inverse for the rescaled two-row boundary-symbol matrix. For $\xi' \neq 0$ this is the ordinary Lopatinskii–Shapiro right inverse. At the pure normal ray $\xi' = 0$, it is the normality of the two independent boundary conditions; loss of rank there would leave a nonzero unconstrained Green boundary jet and is incompatible with a self-adjoint fourth-order realization. Apply the right inverse to $-h^{k_j} B_j u_h$, and lift the resulting four jets by the usual Poisson trace extension $e^{-x_d/h}$ (with cutoffs). The rescaled estimates from (3) are

$$\|v_h\|_{L^2} = O(h^{1/2}), \quad \|D^\alpha v_h\|_{L^2} = O(h^{1/2-|\alpha|}), \quad |\alpha| \leq 4.$$

Lower-order coefficient and cutoff errors are removed successively; the resulting geometric symbol expansion has the same bounds. Equivalently, this is the standard semiclassical right inverse for a normal elliptic boundary trace system, as in the parameter boundary calculus of Agranovich–Vishik [3]. A final smoothing correction makes the boundary conditions exact without changing the estimates. Thus $B_j(u_h + v_h) = 0$, and the last display gives the claimed estimate for Pv_h . $\square$

Lemma 2 (Nonnegativity forces augmented Lopatinskii–Shapiro). *Under the two standing source hypotheses, $(D_s^4 + \Delta^2, B_1, B_2)$ satisfies the Lopatinskii–Shapiro condition at every nonzero tangential covector (σ, ξ') on $(0, S_0) \times \partial\Omega$.*

Proof. At $\sigma = 0$ and $\xi' \neq 0$, this is the ordinary source Lopatinskii–Shapiro condition. Suppose instead that it fails at (x^0, σ, ξ') with $\sigma \neq 0$. By the normal-root formulation of the complementing condition, there is a nonzero exponentially decreasing w on $\mathbb{R}_+$ satisfying

$$(P_0(x^0, \xi', D_t) + \sigma^4)w = 0, \quad B_{j,0}(x^0, \xi', D_t)w(0) = 0. \quad (4)$$

Normalize the packet (2) so that $\|u_h\|_2 = 1 + o(1)$. The frozen equation, coefficient variation on the $\sqrt{h}$ tangential scale, and cutoff counting give

$$\|(P + h^{-4}\sigma^4)u_h\|_2 = O(h^{-7/2}). \quad (5)$$

Let v_h be supplied by Lemma 1, and put $q_h = u_h + v_h$. Then $q_h \in \mathcal{D}(\mathcal{P})$, $\|q_h\|_2 = 1 + o(1)$, and

$$\|(\mathcal{P} + h^{-4}\sigma^4)q_h\|_2 = O(h^{-7/2}), \quad (6)$$

because $h^{-4}\|v_h\|_2 = O(h^{-7/2})$.

Self-adjoint nonnegativity and (6) now imply

$$\begin{aligned} 0 &\leq \langle \mathcal{P}q_h, q_h \rangle \\ &= -h^{-4}\sigma^4\|q_h\|_2^2 + O(h^{-7/2})\|q_h\|_2, \end{aligned}$$

which is negative for small h . This contradiction proves the augmented Lopatinskii–Shapiro condition. $\square$

4 Full answer

Theorem 3 (General-boundary bi-Laplacian spectral inequality). *Assume the boundary operators and the realization $\mathcal{P}$ satisfy the standing hypotheses of Le Rousseau–Zongo [1]: ordinary Lopatinskii–Shapiro, self-adjointness, and nonnegativity. For every nonempty open $\omega \subset \Omega$ there is $C > 0$ such that, for all $\mu \geq 0$,*

$$\|u\|_{L^2(\Omega)} \leq Ce^{C\mu^{1/4}}\|u\|_{L^2(\omega)}, \quad u \in \text{span}\{\phi_j : \mu_j \leq \mu\}.$$

Thus the spectral-inequality question in Section 1.6.2 of arXiv:2109.01521v2 has an affirmative answer.

Proof. Lemma 2 supplies the only additional hypothesis in Zongo–Robbiano [2, Theorem 1.3]. Applying that theorem gives the displayed inequality with the exponent $\mu^{1/4}$. $\square$

5 Verification, scope, and novelty check

Hypothesis match. The source question occurs on page 8 of arXiv:2109.01521v2. The supporting spectral theorem occurs as Theorem 1.3 on page 6 of arXiv:2309.02181v1. Its extra assumption is exactly ordinary Lopatinskii–Shapiro for the augmented operator $D_s^4 + \Delta^2$. Lemma 2 derives that assumption from the source’s self-adjoint nonnegativity.

Sanity checks. The sign is forced: Fourier transformation in s changes D_s^4 into $+\sigma^4$, so a failed augmented condition gives a frozen mode at the negative energy $-\sigma^4$. The packet error is $h^{-7/2}$ versus the main h^{-4} term, leaving the required half-power margin. When $\sigma = 0$, the argument correctly reduces to the source Lopatinskii–Shapiro condition.

Bounded novelty search. On 27 August 2026 we searched the run indexes, arXiv/web results for the exact phrases “spectral inequality”, “general boundary conditions”, “bi-Laplace”, and “Lopatinski–Shapiro”, and the citation trail of arXiv:2109.01521. We found arXiv:2309.02181, which proves the desired estimate with the additional augmented Lopatinskii–Shapiro hypothesis and cites the source paper, but no primary source removing that hypothesis by self-adjoint nonnegativity. This is a bounded search, not a claim of exhaustive novelty.

Human review recommendation. The result is recommended for expert review as a likely valid full solution. The decisive review point is the standard high-frequency right inverse in Lemma 1; a reviewer should check its normality convention against the boundary-operator class in the source. No claim is made about Section 1.6.1 (boundary damping) or Section 1.6.3 (a quantitative interpolation inequality).

References

- [1] J. Le Rousseau and E. Wend-Benedo Zongo, *Stabilization of the damped plate equation under general boundary conditions*, arXiv:2109.01521v2 (2021); J. Èc. polytech. Math. 10 (2023), 1–65.
- [2] E. Wend-Benedo Zongo and L. Robbiano, *Null-controllability for a fourth order parabolic equation under general boundary conditions*, arXiv:2309.02181v1 (2023), especially Theorem 1.3.
- [3] M. S. Agranovich and M. I. Vishik, *Elliptic problems with a parameter and parabolic problems of general type*, Russian Math. Surveys 19:3 (1964), 53–157.